

MathWorks offer tools that help developers minimise the cost of developing a product by enabling testing and verification of the product in the initial stages of design.

Rao adds, "The cost of developing a product increases exponentially when system design errors are found after the implementation or integration of a product. But software testing tools at present are able to assist in finding design errors in the initial development stages, thus reducing the cost that goes into fixing the system errors. They enable early verification of the system so that a designer can quickly find an error and deal with it."

Designers can run the simulation and also move to rapid iteration and

hydraulic system. In real life, if you consider something like an anti-locking brake system, a plant does not consist of just one domain. It has multiple domains within it. In order to verify the controller model for the plant, you have to simulate all domains such as electric, mechanical and hydraulic together with the controller model; you need a multi-domain environment that can handle this type of a simulation," explains Rao.

Physical modelling, for custom domains

Rao says, "We have innovations and investments in physical modelling tools for controller verification that help customers develop their own

building the physical system.

Multi-cores, multi-crores

Faster systems mean faster execution and faster profits, whatever the domain might be. As a result, multi-core processors—chips with more than one processing unit forged on the same die—have gone way beyond supercomputing and entered arenas such as desktop and embedded computing as well. However, to be able to make good use of the multiple processors on-board, a system needs software that makes good use of the chip's parallel computing abilities. This capability should reflect in the modelling tools too.

"At The MathWorks, there is ongoing investment in the area of parallel computing to enable quick and fast simulation, and take advantage of many of the features that newer computers offer such as multi-core systems. We also support computational clusters," says Rao.

Other modelling tools including NI's LabView and IBM's Rational Rhapsody are also ready for the multi-core challenge that is taking the embedded world by storm.

Benefits are hard to ignore!

In the embedded world, delays, cost overruns or errors in a system could lead to similar issues in the products into which these systems are to be fitted. Hence it is not surprising that more and more engineers are adopting a model-driven approach to embedded systems design and development wherein the software and the system are modelled together in sync with the user's requirements.

This approach has proved effective not just in embedded systems but in many other spheres of computing as well. Speed, simplicity, early detection of errors, component reuse—the benefits are many and hard to ignore! ●

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The most acknowledged benefit of modelling tools is in testing. No other approach makes it so easy to avoid and detect errors

rapid prototyping to analyse the possible errors that may creep in after the final integration of the product. What is more, they can do it in a multi-domain environment.

Handling multiple domains easily

Developers can choose between a domain-specific modelling tool and a generic one. Domain-specific tools are applicable when the embedded system is to be applied to a very specific task. However, consider a control system that can be used in a variety of production environments, with mild changes. If a developer had to code separately for each domain, it would be a big time-eater. On the other hand, if you could model a generic system and simply refine it by adding domain-specific information, it would be much easier to deal with multi-domain systems.

"Let us say you have to design a system that controls a plant. The plant could be any system that you are trying to control. It could be an electric system, a mechanical system or a

modelling domains as well." Other modelling tools such as MapleSim are also into cutting-edge physical modelling now.

Physical modelling enables engineers to develop the software and the physical system together. In this approach, the model maps directly to each physical component in the system showing how those components are connected, and the equations of the model are automatically generated to fit this system.

You can create multi-domain physical systems by assembling system-level models that span multiple physical domains and include the control system in a single environment. It is also possible to create reusable models of your physical system with physical ports, in addition to input and output signals.

According to MapleSoft, the maker of MapleSim, Toyota has been one of the earliest industrial companies to embrace this approach to physical modelling—creating a computer-based model of the system to analyse, test, improve and optimise the design before actually